

1(a)(i)	thylakoid(s) ;	1
1(a)(ii)	any four from: up to three descriptions (D) and one / two linked functions (F) D1 stacked / many (A / thylakoids) / large surface area ; F2 to increase light absorption ; D3 (thylakoid) <u>membrane</u> has, (named), pigments / photosystems / electron carriers / enzymes ; F4 (thylakoid membrane role is) light-dependent reaction / photophosphorylation / creates proton gradient / pump protons / chemiosmosis / makes ATP ; D5 (thylakoid) lumen / space ; F6 accumulates H⁺ / high H⁺ concentration ; D7 appear green (under microscope) ; F8 as / so, chlorophyll absorbs red and blue light ;	4
1(b)(i)	Calvin (cycle) ;	1
1(b)(ii)	any two from: 1 no start and end, point / molecules ; 2 all, molecules / intermediates, present all the time ; 3 ribulose biphosphate / RuBP / 5C molecule, is regenerated ; 4 numerical detail ;	2
1(c)(i)	B: oxygen ; D: sugar / hexose / glucose / triose phosphate / carbohydrate / starch ;	2
1(c)(ii)	1 (oxygen linked to) <u>aerobic</u> respiration / <u>oxidative phosphorylation</u> ; 2 (glucose stated as) source / store, of (chemical) energy / ATP / food ; 3 <i>ref. to</i> <u>energy</u> flow through, food chains / food web / ecosystem ;	3